

Evaluating the Performance of Object Detection Models for Accurate Sri Lankan Traffic Sign Detection



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Introduction.

Introduction

- Importance of traffic sign detection in AI and computer vision.
- Relevance to Sri Lankan road systems.
- Comparing YOLOv8, Faster R-CNN, and SSD using mAP and F1-Score.



Research Background.

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- Rising significance of **autonomous vehicles** and **driver assistance systems**
 - Reliable **traffic sign detection** is a key requirement for road safety and autonomous driving
 - The performance of object detection models varies under different conditions, especially in **real-world scenarios**
 - Few comparative studies exist focusing on their performance with **Sri Lankan road signs**
 - This study aims to fill that gap by evaluating **YOLOv8, Faster R-CNN, and SSD** for their accuracy in detecting **Sri Lankan traffic signs**

Problem Statement.



- Traffic sign detection is crucial for autonomous vehicles and road safety.
- Currently, object detection models show varying performance under different conditions.
- There's a lack of comparative studies evaluating their accuracy on Sri Lankan road signs.
- This inconsistency poses a challenge for developing reliable autonomous vehicle technologies.
- This study aims to compare and identify the most accurate and efficient model for Sri Lankan conditions.

Research Question & Hypothesis

Research Question



Research Question: Which object detection model (YOLOv8, Faster R-CNN, or SSD) achieves the highest accuracy in detecting Sri Lankan traffic signs, as measured by mean Average Precision (mAP) and F1-Score?



Alternative Hypothesis: Among the three models YOLOv8 would achieve the highest combined performance based on mAP and F1-Score.



Dependent Variables: Mean Average Precision (mAP) (the accuracy of the models), F1-Score.



Independent Variable: The object detection model used (YOLOv8, Faster R-CNN, SSD).

Methodology



Research Design: Quantitative research.



Data Collection Method: Secondary dataset (Sri Lankan traffic sign dataset).

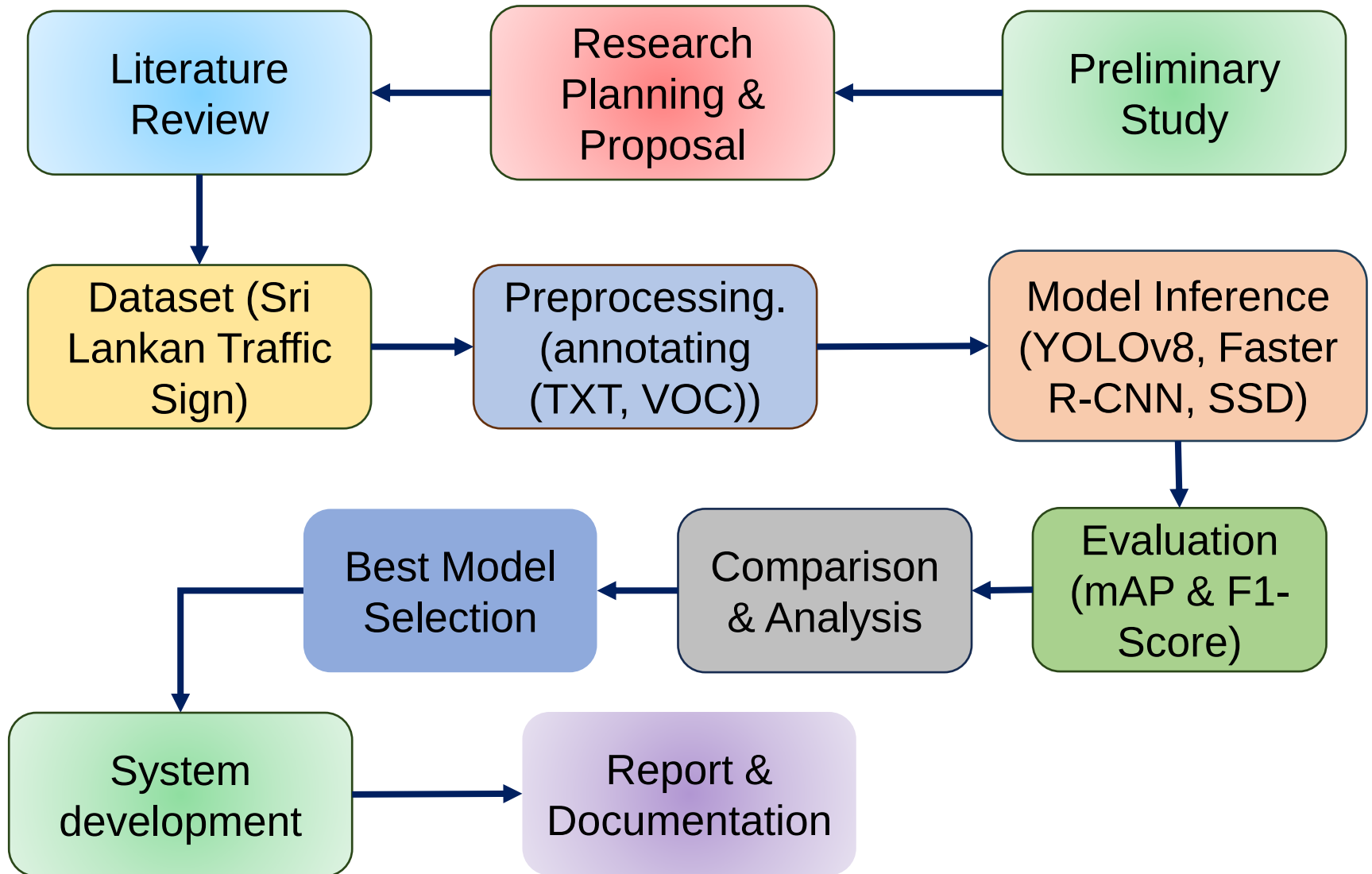


Evaluation Approach: Analyze the performance of YOLOv8, Faster R-CNN, and SSD. Use mean Average Precision (mAP) and F1-Score as the key evaluation metric.



Comparison & Analysis: Compare the models based on mAP and F1 scores. Identify the most effective model for Sri Lankan traffic sign detection.

Top level architecture



Literature Review

Paper: *Performance Comparison of Object Detection Models for Road Sign Detection Under Different Conditions* (Fatima et al., 2024)

Context & Aim

- Compare state-of-the-art detectors (YOLOv8, YOLOv9, RTMDet, Faster R-CNN, RetinaNet) under varied environmental and operational conditions in road sign detection.

Methodology

- Fine-tuned models on large-scale datasets.
- Evaluated using metrics: mean Average Precision (mAP), frames-per-second (FPS), and model loss.
- Included additional training enhancements: data augmentation and hyperparameter tuning.

Results

- YOLOv8 and YOLOv9 achieved highest real-time detection rates.
- RTMDet offered the best trade-off between speed and accuracy.
- Faster R-CNN and RetinaNet delivered higher precision but were slower.

Research Gap

- Analysis conducted on general (non-region-specific) datasets.
- Lacks focused evaluation on the unique characteristics of Sri Lankan traffic signs.

Paper: *A Traffic-Sign Detection Algorithm Based on Improved Sparse R-CNN* (Cao et al., 2021)

Context & Aim

- Enhance Sparse R-CNN for improved detection of traffic signs by optimizing the backbone and attention mechanisms.

Methodology

- Added Res2Nest backbone and Dynamic Instance Interactive Head (DIIH) for richer multi-scale feature representation.
- Evaluated on standard TT100K traffic sign dataset.

Results

- Improved mAP and robustness for small sign detection.
- Backbone enhancements and attention modules enabled more accurate region proposals.

Research Gap

- Higher computational cost may limit real-time applicability.
- Evaluations not performed on datasets representative of Sri Lankan road signs.

Paper: *Evaluating and Benchmarking Object Detection Models for Traffic Sign Detection* (Mishra et al., 2022)

Context & Aim

- Benchmark a variety of object detection models (Faster R-CNN, YOLOv3, SSD, RetinaNet, CenterNet, Cascade R-CNN) using multiple traffic sign datasets.

Methodology

- Employed precision, recall, and mAP across TT100K and GTSD datasets.
- Comparative analysis under diverse real-world conditions including occlusion and small object detection challenges.

Results

- CenterNet and Cascade R-CNN achieved highest overall accuracy, especially for small signs.
- Faster R-CNN and YOLOv3 showed strong baseline performance but were slightly behind top performers.

Research Gap

- Performance degrades in adverse weather and occlusion scenarios.
- No testing on Sri Lankan traffic sign dataset; regional generalization remains untested.

Dataset in brief



The dataset used in this study was sourced from **Roboflow**, a publicly available repository of labelled image datasets for object detection.



It comprises **5,404** images in total, divided into training, validation, and testing sets.



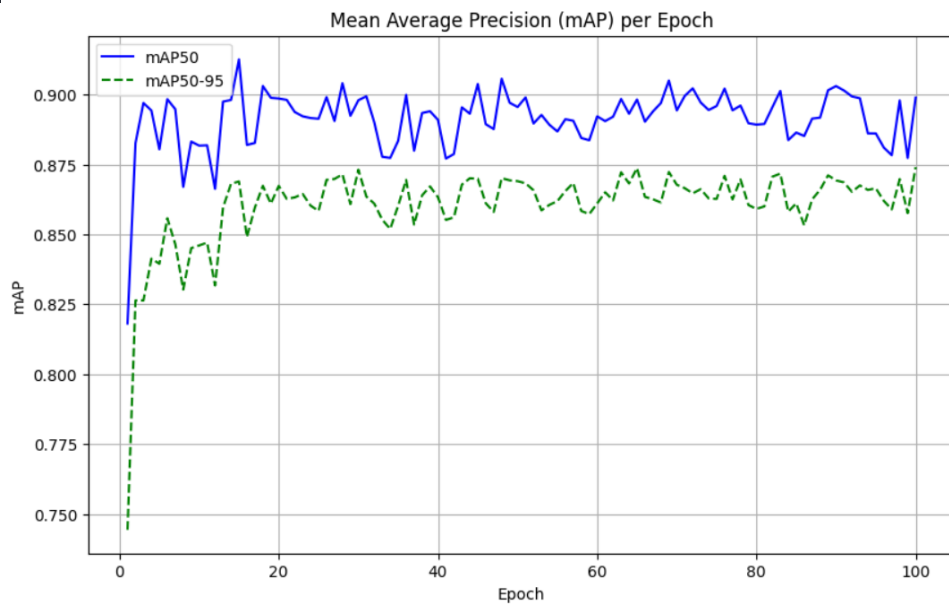
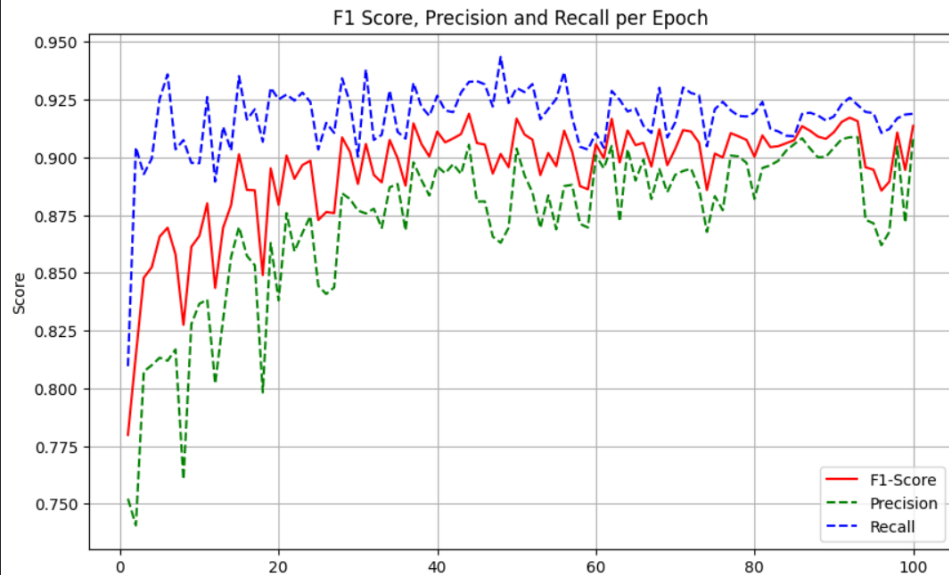
The dataset includes **36** classes of Sri Lankan road signs, encompassing a range of regulatory, warning, and informational signals frequently encountered by drivers.

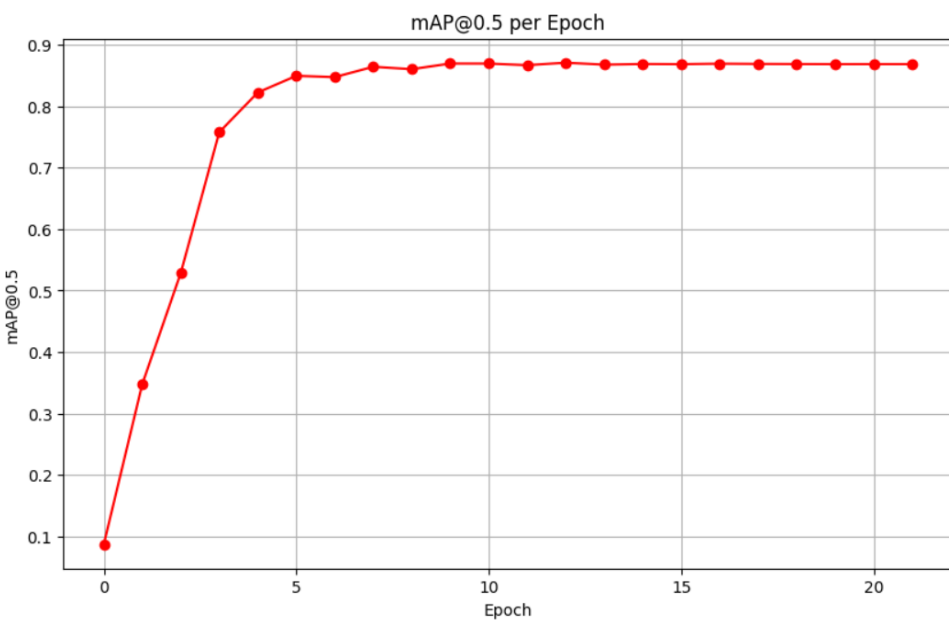
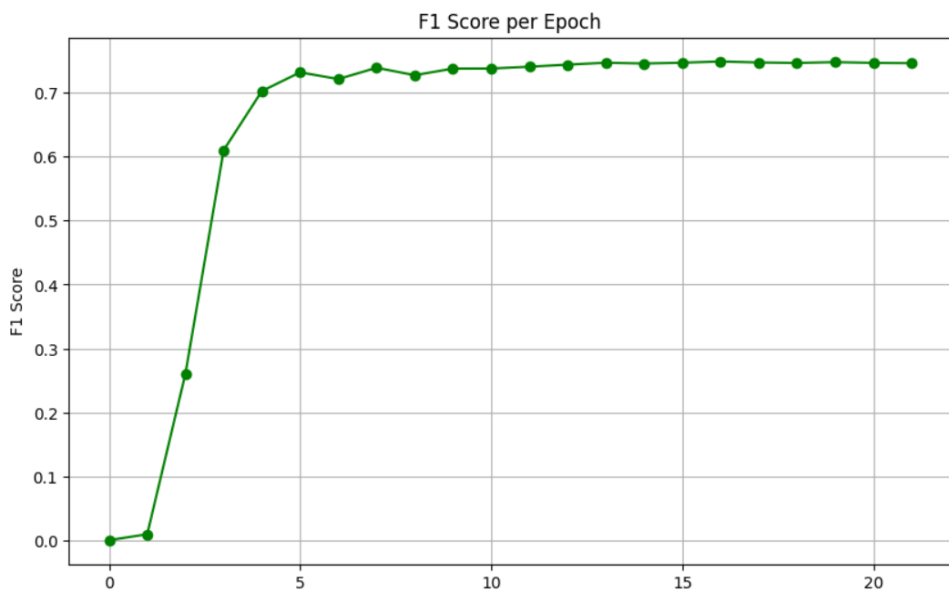


Annotations for each image are provided in **VOC** and **TXT** formats, following standard object detection practices with bounding boxes to aid algorithm training and evaluation.

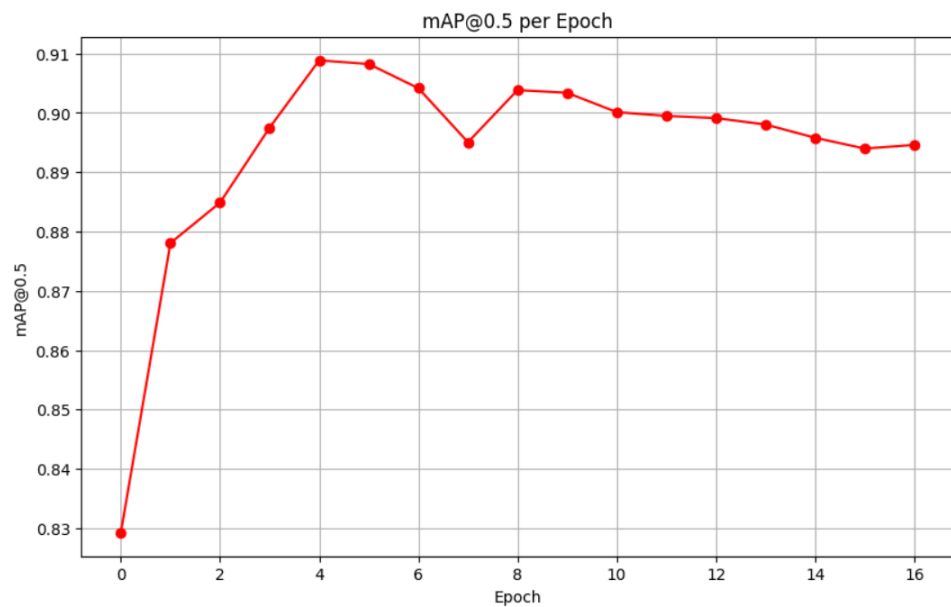
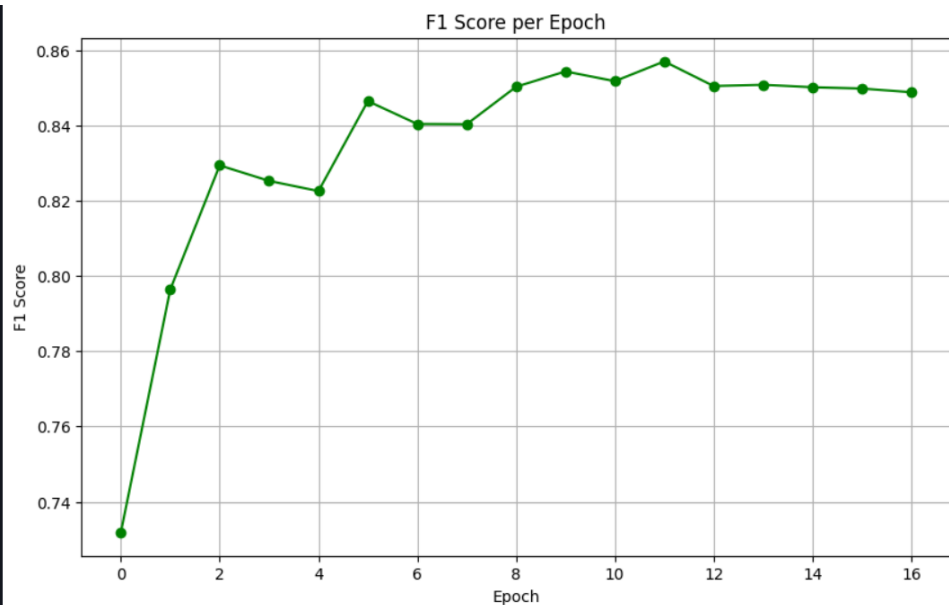
Model Evaluation

YOLOv8



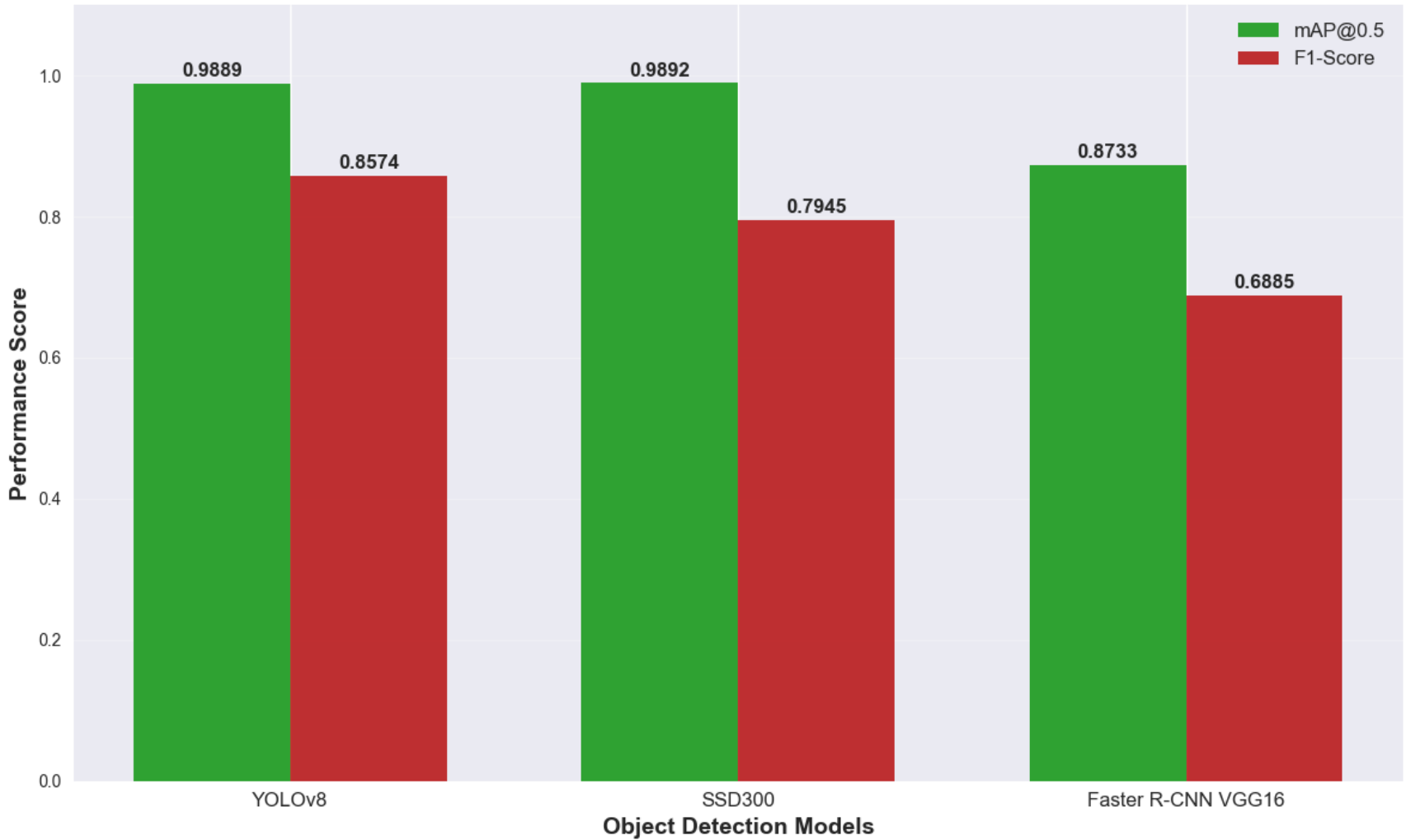


SSD 300



Faster
R-CNN
VGG16

Object Detection Models Performance Comparison mAP@0.5 vs F1-Score

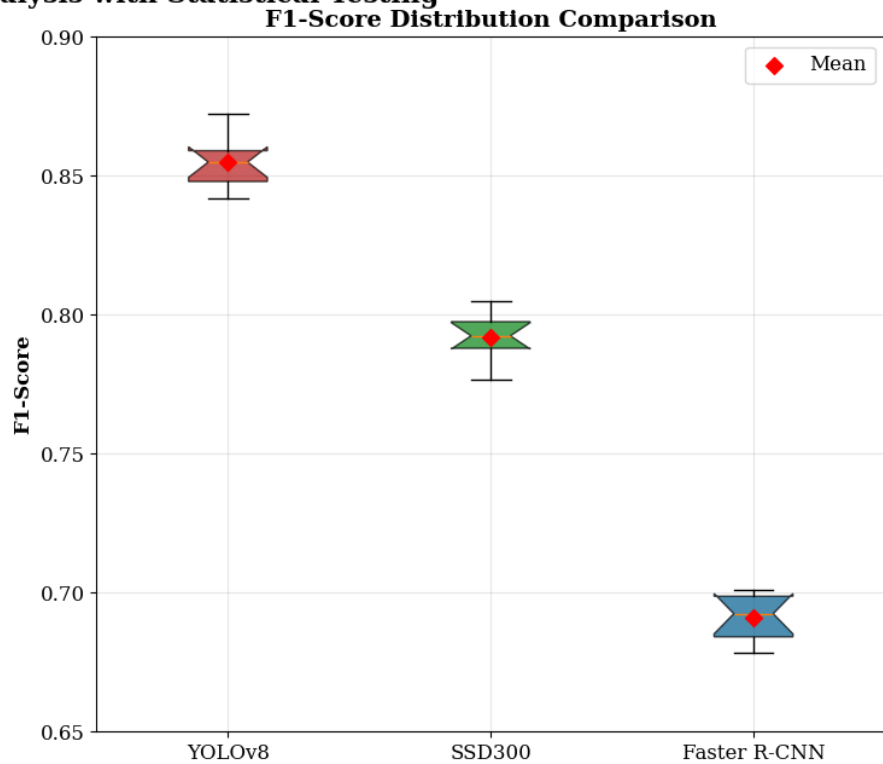
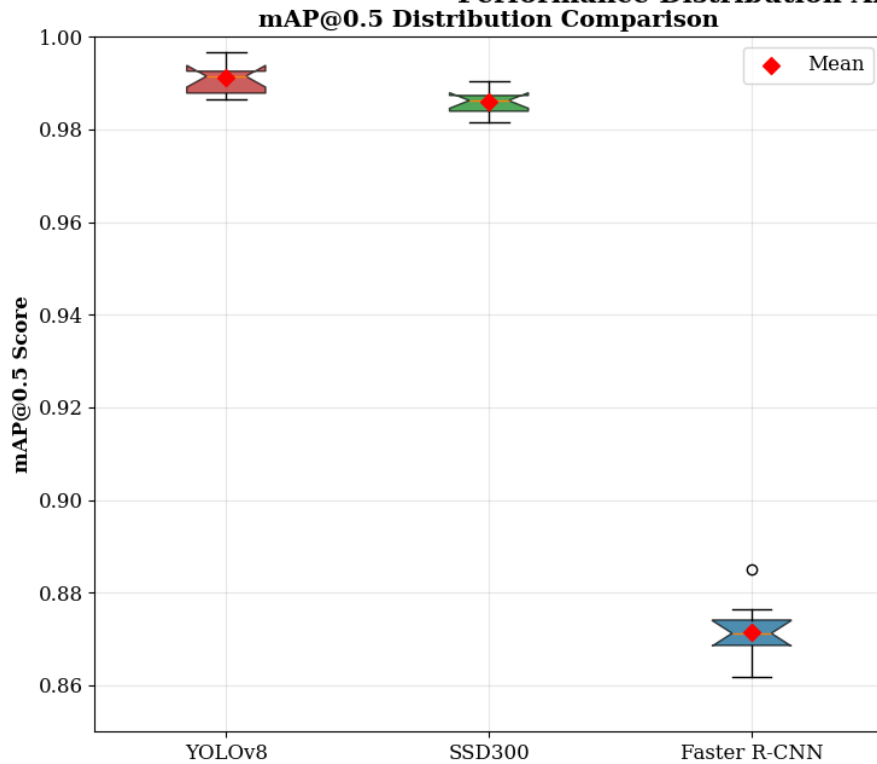


Main Findings

Performance Metrics for Object Detection Models
(Numbers in parentheses indicate ranking)

Model	mAP@0.5 (Rank)	F1-Score (Rank)
YOLOv8	0.9889 (2)	0.8574 (1)
SSD300	0.9892 (1)	0.7945 (2)
Faster R-CNN	0.8733 (3)	0.6885 (3)

Performance Distribution Analysis with Statistical Testing



Demonstration

Limitations

Dataset imbalance across 36 classes

No fine-tuning of pre-trained models

Limited testing under varied environmental conditions

Dataset not covering all the signs across the country

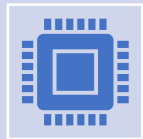
Future work



Develop balanced and larger dataset



Evaluate performance under varied weather & lighting conditions



Optimize for deployment on edge devices (Jetson, Raspberry Pi)



Fine-tune models for local dataset

Conclusion

Conclusion

The research identified YOLOv8 as the best-performing model for Sri Lankan traffic sign detection.

Hypothesis testing confirmed YOLOv8's superior performance over the other models.

A real-time detection system using YOLOv8 and Streamlit was successfully developed.

The study supports the application of AI in road safety and intelligent transportation systems in Sri Lanka.

Reference

- Fatima, Z. T. M. M. H. V. R. R. T. a. R. R., 2024. Performance Comparison of Object Detection Models for Road Sign Detection Under Different Conditions. International Journal of Advanced Computer Science and Applications, 15(12).)
- (Jinghao Cao, J. Z. X. J., 2021. A Traffic-Sign Detection Algorithm Based on Improved Sparse R-cnn. IEEE Access.)
- (Mishra, A. K. A. M. S. A. K. Z. J. S. S. a. T. A., 2022. Evaluating and Bench-marking Object Detection Models for Traffic Sign and Traffic Light Datasets, s.l.: Computer Vision Foundation.)



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Thank you

Questions?

